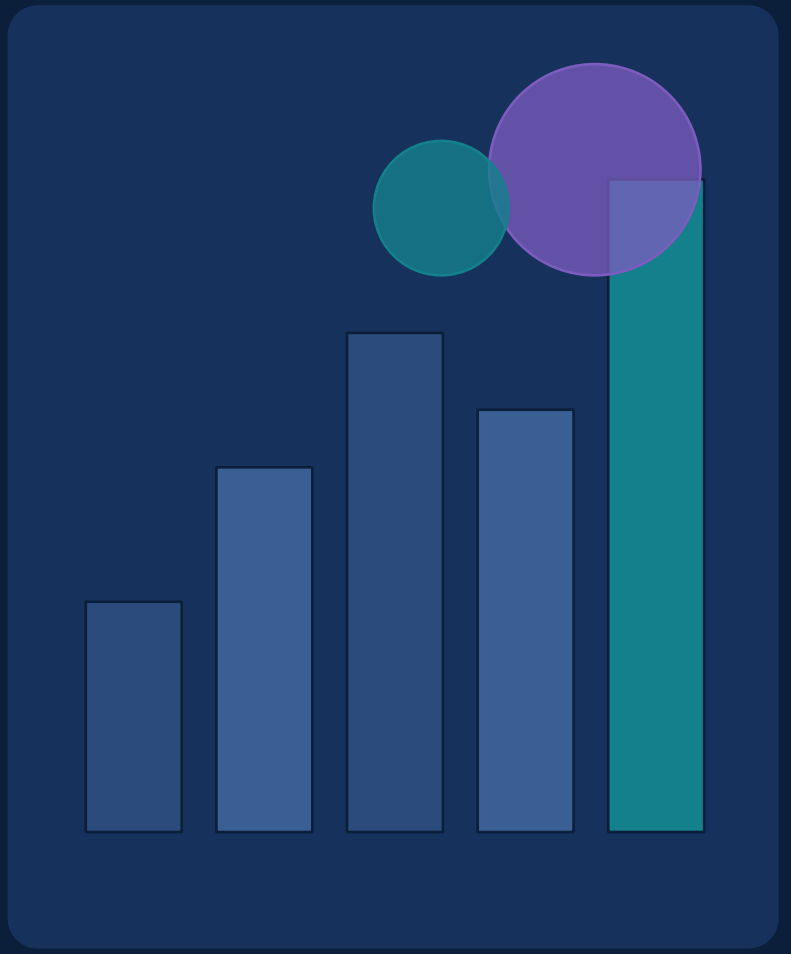
EXECUTIVE REVIEW

Customer Shopping Behavior Analysis

Data-driven insights for customer growth and retail strategy



[Presenter name] | [Organization] | [Presentation date]

Executive summary

A loyal, evenly-spending customer base; growth levers are retention, subscriptions and smarter discounts

3,900	$59.76	3.75	$233,081
Purchase records	Average purchase amount	Average review rating	Total revenue

1	**Retention**	80% of customers are Loyal; only 83 are New — protect the base and grow the pipeline
2	**Subscriptions**	27% subscribe, yet average spend is about the same ($59.49 vs $59.87) — test the value of benefits
3	**Product positioning**	Top-rated items score 3.78–3.86; pair ratings with sales volume when choosing campaigns
4	**Discount management**	Hat and Sneakers are discounted on ~50% of purchases — balance sales lift with margin

Source: Customer Shopping Behavior Analysis report (3,900 purchase records). Total revenue = sum of the report's revenue tables ($233,081).

Business objective and data

Four business questions, answered from 3,900 purchase records

Q1 Who spends, and how much?

Q2 Which customers are loyal, returning or new?

Q3 How do subscriptions, discounts and shipping relate to spend?

Q4 Which products and categories perform best?

3,900
Rows (purchases)

18
Columns

25 / 4
Products / categories

37
Missing ratings, imputed

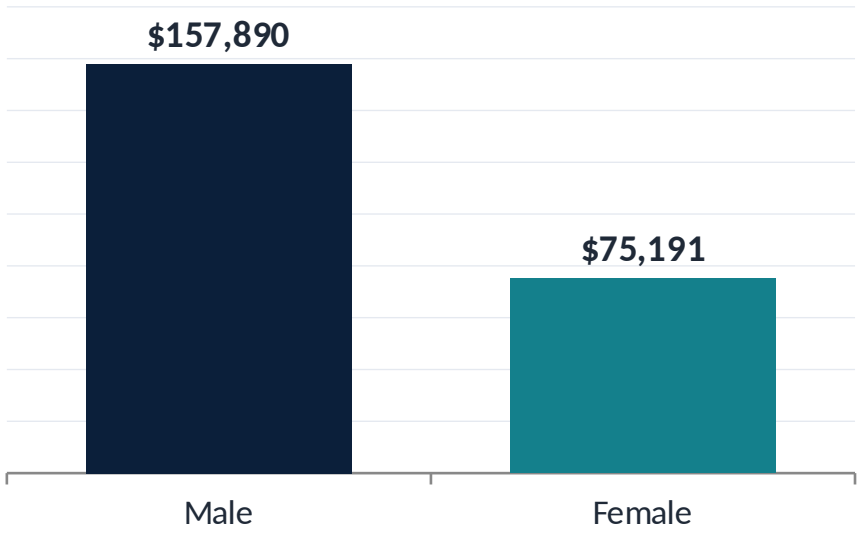
Demographics	Age, Gender, Location, Subscription Status
Purchase details	Item, Category, Amount, Season, Size, Color
Shopping behavior	Discount, Promo Code, Previous Purchases, Frequency, Rating, Shipping Type

Note: each row is one customer ID, so 'customers' and 'purchase records' are both 3,900. 37 missing review ratings were imputed with the category median. Python and SQL steps are in the full report.

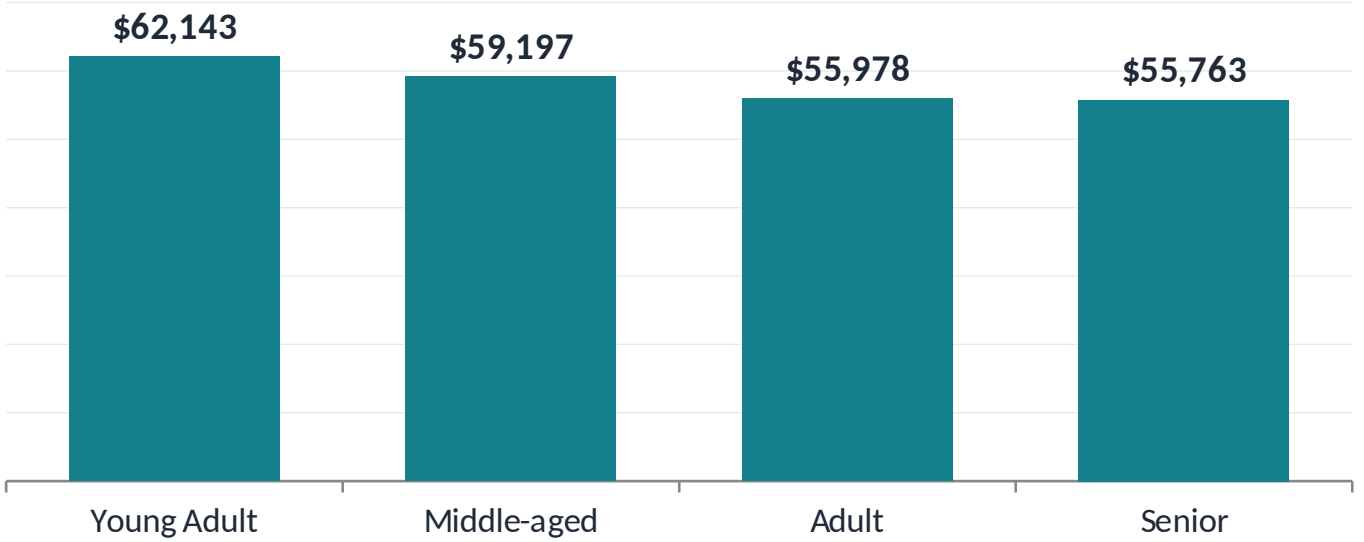
Customer spending and revenue

Revenue is concentrated in male customers and led modestly by Young Adults

Revenue by gender (USD)



Revenue by age group (USD)



Gender mix drives the gap

Males: ~68% of customers and ~68% of revenue (derived)

Age gap is modest

Young Adult revenue is ~11% above Senior — a test, not a certainty

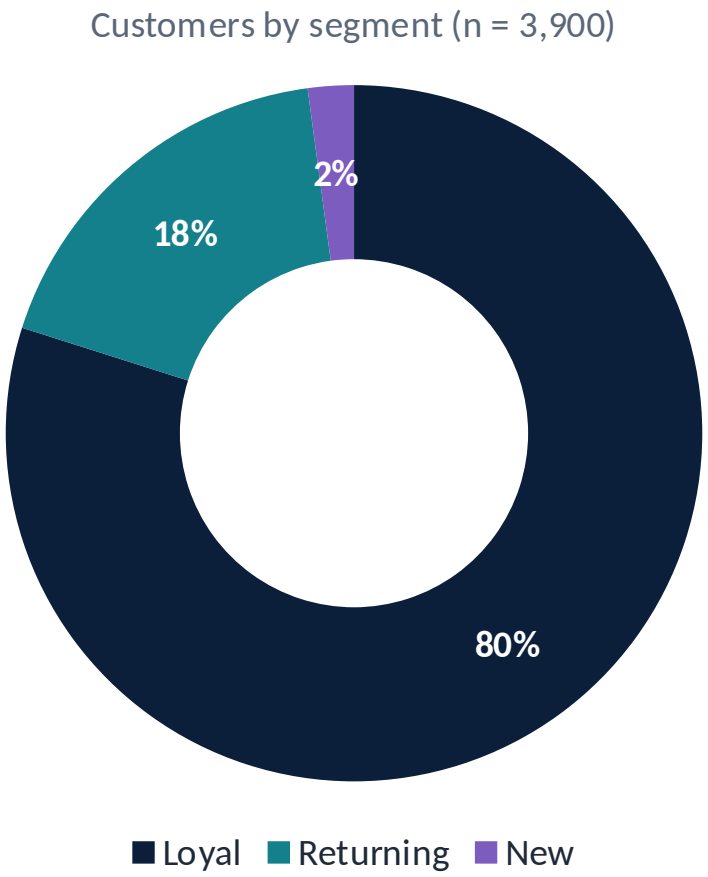
Revenue is not profit

Margin is not in the data; validate before shifting spend

Source: report SQL results. Customer counts by gender come from the Python summary (Male = 2,652). Age-group bin boundaries are not stated in the report.

Customer segmentation and loyalty

Four in five customers are Loyal; only 83 are New



Loyal	**3,116**	79.9%
Returning	**701**	18.0%
New	**83**	2.1%

Opportunity (recommendation)

Convert New to Returning with a second-purchase offer, and reward Returning customers to move them toward Loyal.

To investigate: why the New group is so small.

Segments per the report's purchase-history rules (thresholds not stated in the report). Percentages are derived from the reported counts.

Subscription performance

Subscribers spend about the same; non-subscribers just outnumber them

Subscribers

Customers

1,053

Average spend

$59.49

Total revenue

$62,645

Repeat buyers (>5 purchases)

958 (91.0%)

Non-subscribers

Customers

2,847

Average spend

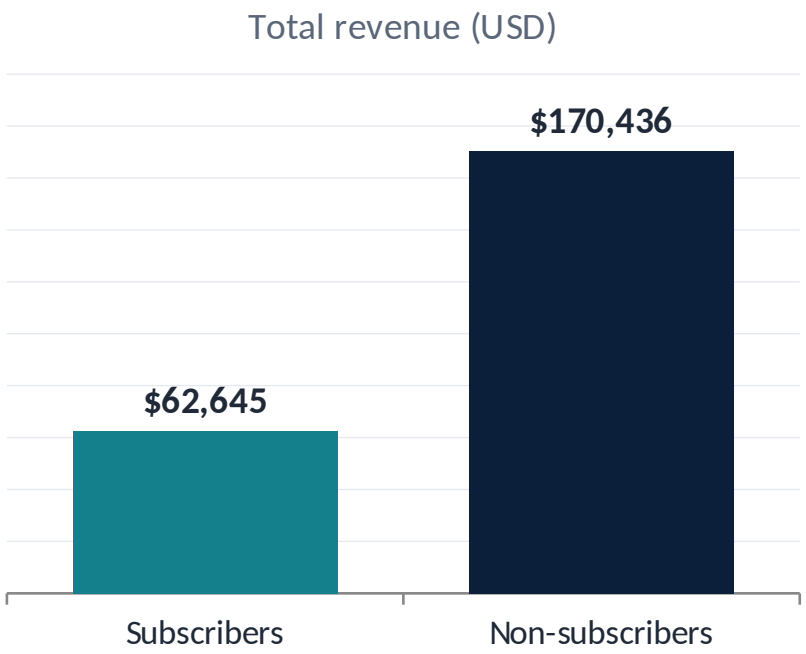
$59.87

Total revenue

$170,436

Repeat buyers (>5 purchases)

2,518 (88.4%)

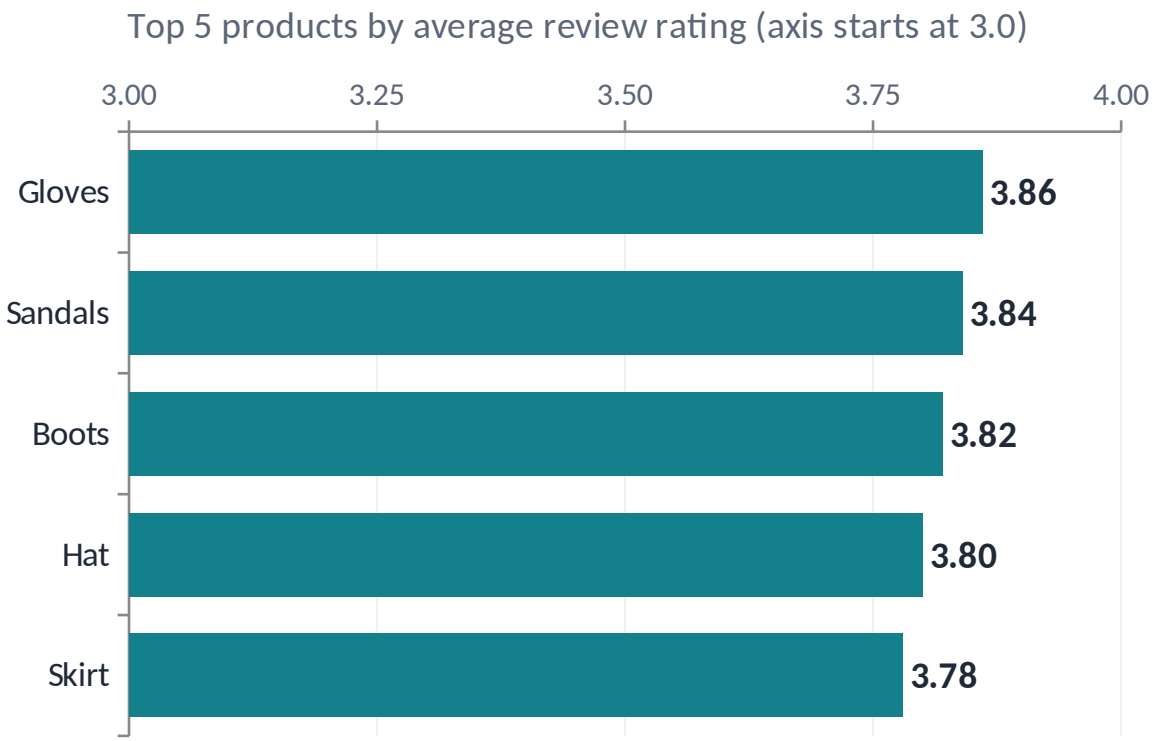


Non-subscribers deliver ~73% of revenue because they are ~73% of customers. Subscription is not shown to drive spend: test benefits and track conversion, retention and incremental revenue.

Repeat-buyer rates are derived: 958 of 1,053 subscribers and 2,518 of 2,847 non-subscribers have more than 5 purchases. Comparison is descriptive, not causal.

Product performance and preferences

Top ratings are tightly bunched; ratings and sales volume tell different stories



Best-selling item in each category (orders)

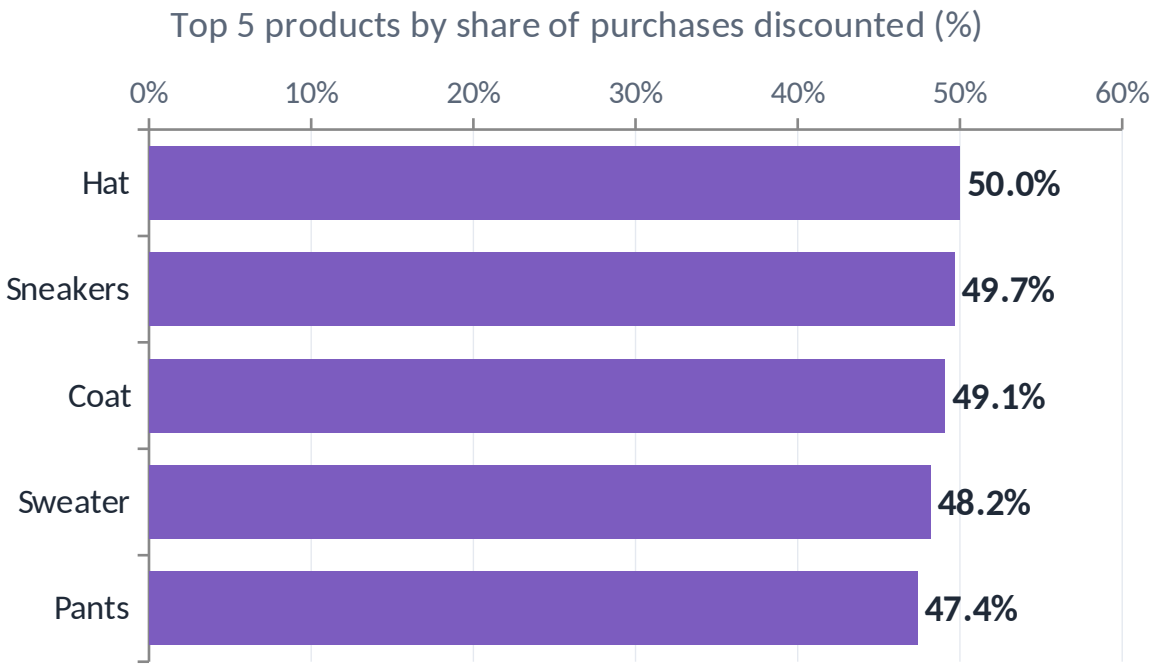
Accessories	**Jewelry**	171
Clothing	**Blouse**	171
Footwear	**Sandals**	160
Outerwear	**Jacket**	163

Sandals rank in both the top-5 ratings and as the top Footwear seller — prioritize products strong on both measures.

Ratings are averages on a 1–5 scale; chart axis starts at 3.0. Best sellers are by order count (top item per category). The report's top-3 table is truncated for Outerwear (only Jacket and Coat visible).

Discounts, shipping and purchase behavior

Discount reliance is high on a few items; shipping type barely moves average spend



Average purchase amount by shipping type

$58.46
Standard shipping

$60.48
Express shipping

Express is $2.02 (3.5%) higher — a pattern, not a proven cause.

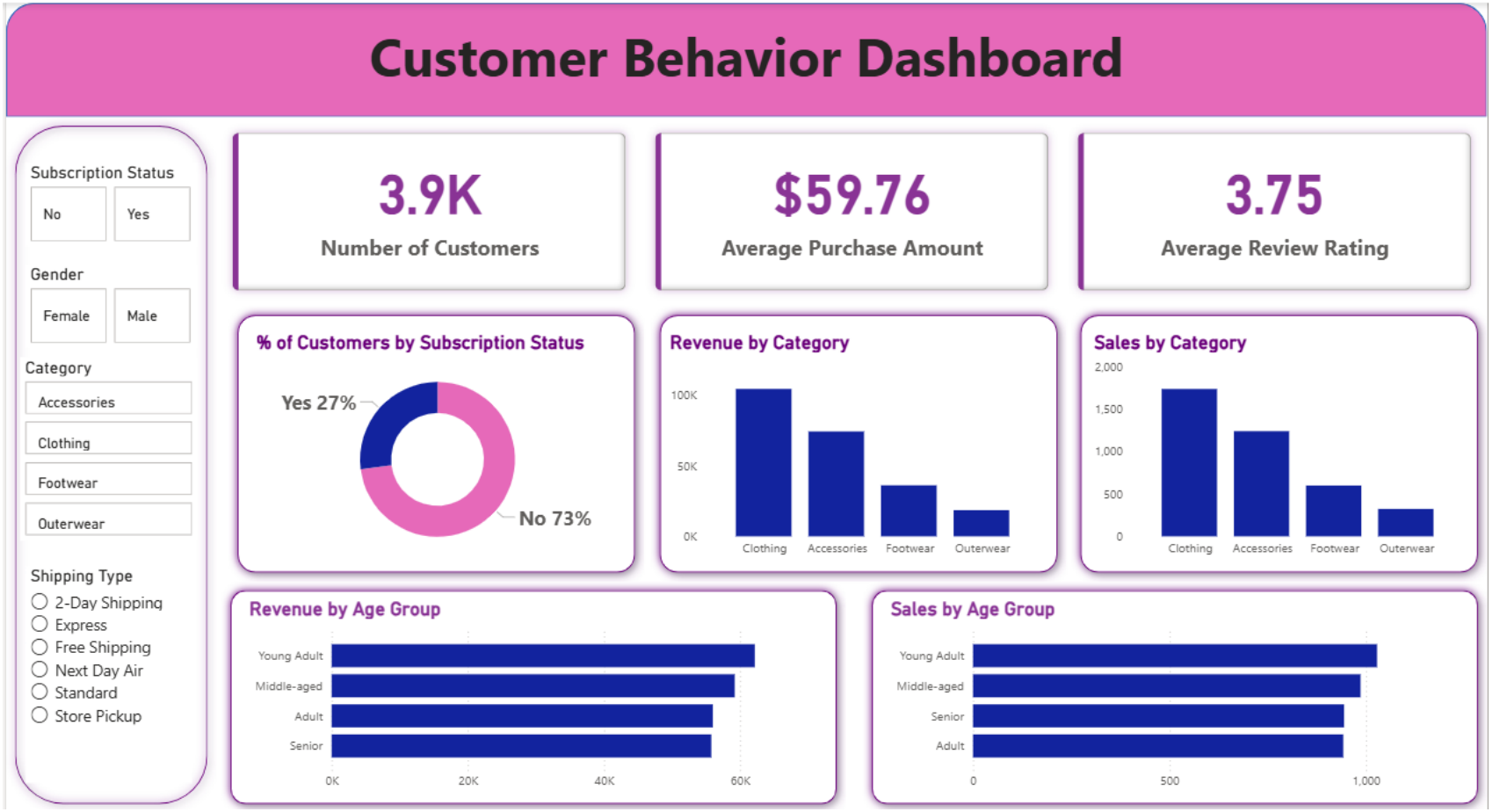
Opportunity: optimize discount depth on the most-discounted items; use shipping choice as a targeting segment.

Needed for stronger decisions: Margin | Shipping cost | Conversion rate

Source: report SQL results. 43% overall discount rate derived from the Python summary (2,223 of 3,900 purchases without a discount). The data has six shipping types; only Standard and Express were compared.

Power BI customer behavior dashboard

One interactive view for customers, spend, ratings, categories and age groups



KPI cards

Customers, average purchase, average rating

Slicers

Filter by subscription, gender, category, shipping type

Breakdowns

Category and age group, by revenue and by sales

Screenshot of the Power BI dashboard from the report. Category and age-group values are read from the report's own charts.

Key business opportunities

Four areas where the data points to action

1. Retention and loyalty

Observation: 3,116 Loyal, 701 Returning, 83 New

Implication: Few New customers may limit the pipeline (hypothesis)

Action: Pilot a second-purchase offer; track New-to-Returning conversion

2. Subscription conversion

Observation: 27% subscribe; avg spend $59.49 vs $59.87

Implication: Subscribers do not spend more today; value must come from retention (to test)

Action: A/B test benefits; measure conversion, retention, incremental revenue

3. Discount and margin

Observation: Hat 50.0%, Sneakers 49.7% of purchases discounted

Implication: Some discounts may not be needed (hypothesis); margin unmeasured

Action: Add margin data; test shallower discounts on top items

4. Product marketing and targeting

Observation: Ratings 3.78–3.86; Young Adults lead revenue ($62,143)

Implication: Ratings and volume differ; age gaps are modest

Action: Feature items strong on both (e.g., Sandals); test Young Adult campaigns

Observations come from the report; implications are hypotheses; actions are recommendations.

Strategic recommendations and KPIs

A prioritized plan with measurable KPIs

1 New-to-returning conversion

% of New customers making a repeat purchase in a set window

No baseline in report

2 Repeat-purchase rate

% of customers with more than one purchase in a period

Report: 3,476 customers with >5 purchases

3 Subscription conversion

% of non-subscribers who subscribe in a period

Report: 27% subscribe today

4 Revenue and margin by segment

Revenue and contribution margin per customer segment

Revenue only; no margin

5 Discount conversion and profit

Conversion lift and margin on discounted vs full-price sales

No baseline in report

6 Product conversion and rating

Conversion rate and average rating per product

Report: avg rating 3.75; no conversion

Suggested KPI definitions only: targets to be set after baselines are measured; nothing here is an existing result.

Conclusion and next steps

Three key findings

1. **A loyal base**
   80% of customers are Loyal; only 2% are New.
2. **Spend is uniform**
   ~$60 average purchase whether subscribed or not; non-subscribers lead revenue by volume.
3. **Modest differences**
   Young Adults lead revenue by ~11% over Seniors; ratings bunch at 3.78–3.86.

Top three actions

1. **Pilot New-to-Returning**
   Second-purchase offer; track conversion.
2. **Test subscription benefits**
   A/B test; measure conversion, retention, incremental revenue.
3. **Add margin and conversion data**
   Then review discount policy on top-discounted items.

Takeaway: use customer data to retain the base, prove the value of subscriptions and discounts, and measure profit, not just revenue.